

Network Security Scanner

Project Overview

The Network Security Scanner is a cybersecurity project developed to identify devices connected to a network and evaluate their security posture. It scans IP addresses, detects live hosts, identifies open TCP/UDP ports, discovers running services, and performs basic operating system detection. This information helps administrators understand network exposure and reduce security risks.

Objectives

- Discover active devices on the network.
- Detect open ports and services.
- Identify possible security weaknesses.
- Generate reports for analysis and remediation.

Tools & Technologies

Nmap, TCP/IP, Networking Fundamentals, Linux, Command Line.

Learning Outcomes

Improved knowledge of network reconnaissance, port scanning, service enumeration, security assessment, and ethical cybersecurity practices.